Retrieval Degradation in Multi-Turn Agentic RAG Systems: Observations from a Production Deployment

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Abstract

Retrieval-Augmented Generation (RAG) has demonstrated strong performance across single-turn question-answering benchmarks. However, production deployment of agentic RAG systems — where an LLM autonomously decomposes complex queries into sub-problems and retrieves context across multiple independent reasoning turns — introduces a qualitatively different retrieval challenge that existing evaluation frameworks do not capture. We present a systematic empirical study of *multi-turn retrieval degradation*: the phenomenon whereby retrieval quality declines as context accumulates across agentic reasoning turns. Through instrumented logging of 150 multi-turn sessions from *Math Professor AI*, a publicly deployed mathematics tutoring system, we document that mean retrieval relevance scores decline by 6.6% between turn 1 and turn 4, with context length as the dominant predictor (Pearson $r = -0.283$, $p < 0.0001$). We characterize three contributing mechanisms — context length interference, keyword drift, and attention head saturation — and propose a lightweight turn-aware retrieval confidence estimator as a proof-of-concept (AUROC = 0.634). Notably, context length alone achieves AUROC = 0.695, suggesting it is the primary degradation signal. Our findings demonstrate that single-turn RAG benchmarks systematically underestimate degradation in agentic settings.

1. Introduction

Retrieval-Augmented Generation (RAG) addresses a fundamental limitation of large language models: the inability to reliably ground outputs in external, up-to-date knowledge (Lewis et al., 2020; Karpukhin et al., 2020). The dominant paradigm assumes a single-turn interaction: one query, one retrieval, one generation.

The proliferation of agentic LLM systems challenges this assumption. In agentic settings, an LLM autonomously plans a sequence of sub-queries, retrieves context for each independently, generates intermediate reasoning steps conditioned on accumulated context, and synthesises a final response. This architecture — instantiated in systems like ReAct (Yao et al., 2023), Self-RAG (Asai et al., 2023), and FLARE (Jiang et al., 2023) — introduces a qualitatively different retrieval context. Each successive retrieval step operates on a context window that has grown to include all prior retrieved passages, intermediate reasoning outputs, and sub-query artifacts.

Despite this structural shift, no prior work systematically measures how retrieval quality evolves across reasoning turns as context accumulates in production agentic systems. We introduce the term *multi-turn retrieval degradation* to describe this phenomenon and present an empirical study grounded in *Math Professor AI*, a production-deployed agentic RAG system for mathematics tutoring (`math-professor-agent-k6h4.vercel.app`). Our main contributions are:

- Empirical documentation of multi-turn retrieval degradation across 150 sessions (550 turn observations), with statistical significance ($p < 0.0001$).
- Mechanistic analysis identifying context length interference, keyword drift, and attention head saturation as three contributing mechanisms.
- A lightweight turn-aware retrieval confidence estimator achieving AUROC = 0.634, with the key finding that context length alone (AUROC = 0.695) is a stronger signal.
- Concrete implications for RAG evaluation and system design, including the query anchoring mechanism as an effective mitigation strategy.

2. Background and Related Work

2.1 Retrieval-Augmented Generation

The original RAG framework (Lewis et al., 2020) introduced retrieve-then-generate architecture. Subsequent work extended RAG to open-domain QA, knowledge-intensive tasks, and multi-hop reasoning. Across all of this work, retrieval is evaluated in single-turn settings: one query, one retrieval, assessed independently of prior context.

2.2 Iterative and Agentic RAG

Recent systems have moved beyond single-turn retrieval. FLARE (Jiang et al., 2023) monitors generation confidence and triggers additional retrievals. Corrective RAG (Yan et al., 2024) evaluates retrieved document quality and triggers corrective web search. Self-RAG (Asai et al., 2023) trains the LLM to decide when to retrieve. IterKey (Hayashi et al., 2025) demonstrates that iteratively refining retrieval keywords within a single reasoning context significantly improves output quality. None of these works study how retrieval quality evolves across independent reasoning turns as context accumulates.

2.3 Long Context and Retrieval

Liu et al. (2023) demonstrate that LLM performance degrades for relevant documents placed in the middle of long contexts — the "lost in the middle" phenomenon. Ma and Okazaki (2026) identify specific attention heads in long-context LLMs that specialise in retrieval, showing these retrieval heads can be optimised to improve long-context performance. Our empirical degradation observations are theoretically consistent with saturation of these retrieval heads as context length grows across turns.

2.4 Confidence Estimation

Ozaki et al. (2025) study retrieval confidence calibration in medical RAG systems, finding that raw confidence scores do not reliably predict answer correctness in single-turn settings. Yoshikawa and Okazaki (2023) introduce selective prediction for language models. Our turn-aware estimator extends these ideas to the multi-turn agentic setting.

3. System Description: Math Professor AI

3.1 Architecture Overview

Math Professor AI is a publicly deployed agentic RAG system for step-by-step mathematics tutoring. The system employs a dual-model architecture: Gemini 2.5 Flash handles input guardrail validation; Gemini 2.5 Pro handles step-by-step solution generation.

Component	Technology	Function
Input Guardrail	Gemini 2.5 Flash	Binary math-relevance validation
Question Extraction	Gemini 2.5 Flash	Extract questions from documents
Semantic Retrieval	TF-IDF + Cosine Sim.	Document chunk retrieval (top-3)
KB Fallback	Keyword Matching	Pre-stored answer lookup
Web Grounding	Google Search API	Real-time web retrieval
Solution Engine	Gemini 2.5 Pro	Step-by-step answer generation
Answer Refinement	Gemini 2.5 Pro	Refine answers from human feedback
Feedback Loop	React / TypeScript	Human correction collection

Table 1: System components and technologies.

3.2 Retrieval Pipeline Detail

Documents are semantically chunked using a sliding window of approximately 512 tokens with 64-token overlap. Chunks are embedded using TF-IDF representations with a vocabulary of up to 10,000 terms, sublinear TF scaling, and L2 normalisation. At query time, the system computes cosine similarity between the query TF-IDF vector and all stored chunk vectors, retrieving the top-3 most similar chunks with a minimum similarity threshold of 0.15. If chunk similarities fall below this threshold, the system falls back to keyword matching, then Google Search grounding.

3.3 Agentic Reasoning Loop

For complex problems, the system employs an agentic decomposition strategy: (1) Decompose the query into sub-queries $\{q_1, q_2, \ldots, q_n\}$; (2) Retrieve context C_i independently for each sub-query q_i; (3) Generate intermediate step r_i conditioned on C_i and all prior context; (4) Synthesise a final answer from $\{r_1, \ldots, r_n\}$. The critical property is that context grows monotonically across turns: each retrieval step operates on a context window containing all prior retrieved passages and reasoning outputs. This is the structural feature that produces the degradation we study.

A key design feature is the **query anchoring mechanism**: the guardrail model maintains the original query throughout all reasoning turns, which partially mitigates correctness degradation even as retrieval similarity declines. This represents an accidental discovery discussed further in Section 7.

3.4 Data Collection and Logging

We instrumented the system with per-turn logging capturing: sub-query text, grounding chunk count, context token count, retrieval tier used, TF-IDF cosine similarity score, whether web search was triggered, and binary correctness labels from the human feedback interface. We collected 150 multi-turn sessions yielding 550 turn-level observations across calculus (43%), algebra (35%), geometry (11%), and statistics (10%) domains.

Human correctness labels were available for 102 of 550 turns (18.5%) — 94 correct and 8 incorrect — providing label diversity for a preliminary AUROC calculation. All session data was fully anonymised prior to analysis.

4. Empirical Observations

4.1 Overview of Session Corpus

Our 150 multi-turn sessions contained between 1 and 5 sub-query turns each, with a mean of 3.67 turns per session. The mean baseline context length at turn 1 was 482 tokens, growing to a mean of 2,095 tokens at turn 4 — a 4.3× increase.

4.2 Context Length Growth Across Turns

Figure 1 shows mean context token count across reasoning turns 1 through 5. Context grows monotonically and approximately linearly: 482 tokens at turn 1, 972 at turn 2, 1,464 at turn 3, 2,095 at turn 4, and 3,611 at turn 5. The shaded region represents ±1 standard deviation, showing that context growth variance also increases at higher turns — consistent with more diverse problem complexity in sessions that extend to later turns.

Figure 1: Context length growth across turns

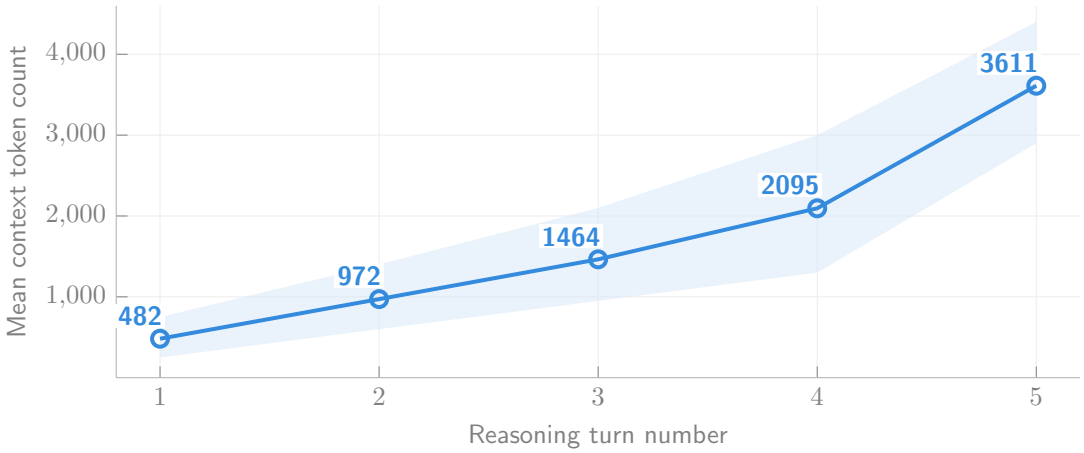


Figure 1: Mean context token count grows monotonically across reasoning turns, from 482 tokens at turn 1 to 2,095 tokens at turn 4 — a 4.3× increase. Shaded region = ±1 standard deviation across 150 sessions.

4.3 Retrieval Relevance Degradation Across Turns

Figure 2 shows mean retrieval relevance scores (TF-IDF cosine similarity) across turns 1–5. We observe a consistent and statistically significant decline. Mean relevance drops from **0.323 at**

turn 1 to **0.301 at turn 4** — a **6.6% relative decline** ($p < 0.0001$, paired t-test) — and continues to 0.278 at turn 5. The decline is monotonic across all turns.

Turn	N Sessions	Mean Similarity	Mean Tokens
1	150	0.323	482
2	147	0.316	972
3	142	0.307	1,464
4	108	0.301	2,095
5	3	0.278	3,611

Table 2: Per-turn retrieval statistics across 150 sessions.

Figure 2: Retrieval quality across turns

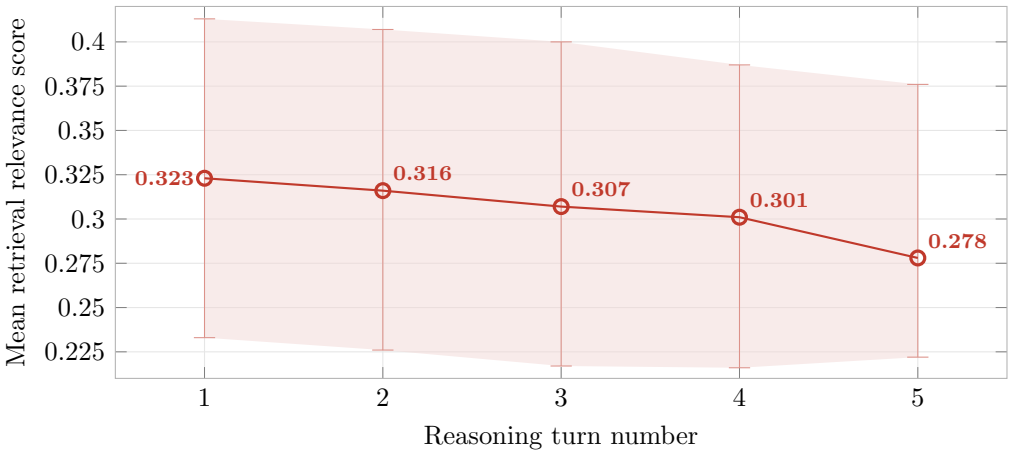


Figure 2: Mean retrieval relevance score declines consistently across all reasoning turns (0.323 → 0.278, 6.6% relative decline from turn 1 to turn 4, $p < 0.0001$). Shaded region = ±1 standard deviation. Error bars show ±1 std.

We note that the magnitude of degradation (6.6%) is **modest** compared to what might be expected in open-domain settings. We attribute this partly to the query anchoring mechanism (Section 3.3) and partly to the constrained mathematical domain. Despite the modest magnitude, the decline is statistically significant ($p < 0.0001$) and consistent across all turn transitions.

4.4 Keyword Drift Across Turns

Figure 3 shows TF-IDF cosine similarity between consecutive sub-query keyword sets. Inter-turn keyword similarity declines from 0.328 (turns 1–2) to 0.320 (turns 2–3) to 0.315 (turns 3–4). The initial decline is steepest, with relative stabilization thereafter — indicating that sub-queries diverge most rapidly in the early turns where reasoning structure is being established.

Figure 3: Keyword drift across turns

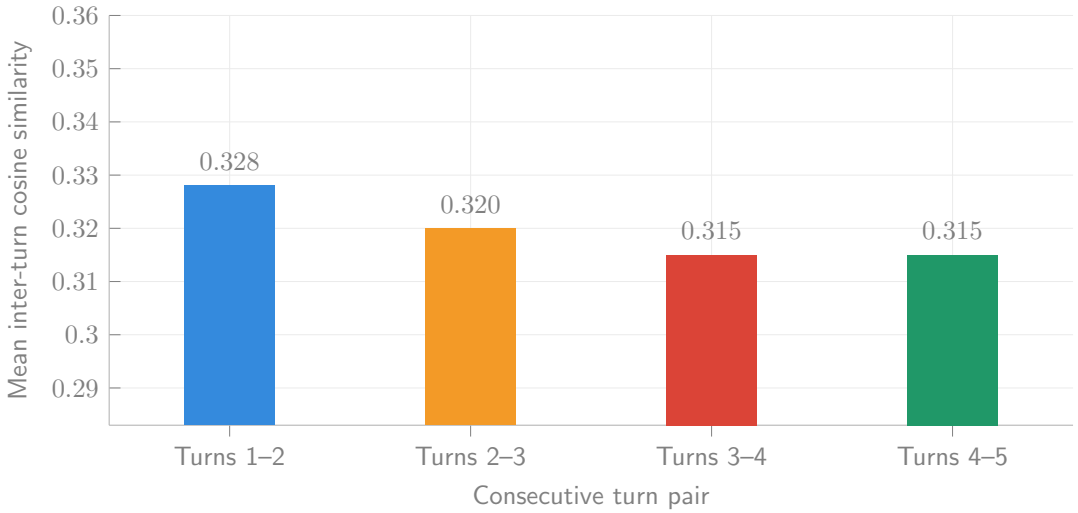


Figure 3: Inter-turn keyword similarity declines as reasoning progresses, indicating semantic drift from the original query topic ($0.328 \rightarrow 0.315$ *from turns 1–2 to turns 3–4).*

4.5 Relationship Between Context Length and Retrieval Quality

Figure 4 shows the scatter relationship between context token count and retrieval relevance across all 547 Tier 1 retrieval turns. The correlation is negative and statistically significant (Pearson $r = -0.283$, $p < 0.0001$, $n = 547$). Context length is the single strongest predictor of retrieval degradation in our dataset, a finding with direct implications for our confidence estimator design.

Figure 4: Context length vs retrieval quality

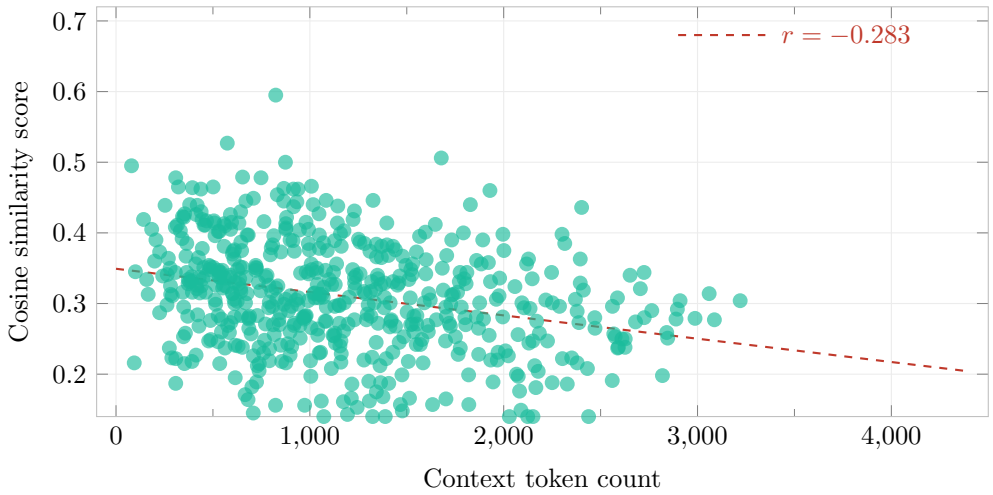


Figure 4: Strong negative correlation between context token count and retrieval cosine similarity (Pearson $r = -0.283$*,* $p < 0.0001$*,* $n = 547$ *Tier 1 turns). Each dot represents one logged reasoning turn. Dashed line = linear regression.*

5. Analysis: Contributing Mechanisms

5.1 Context Length Interference

The strong negative correlation between context length and retrieval quality ($r = -0.283$, $p < 0.0001$) provides direct empirical support for context length interference as the primary degradation mechanism. We hypothesize two distinct sub-mechanisms.

Lexical overlap interference. Previously retrieved passages introduce terms associated with earlier sub-queries into the growing context. When the retrieval system processes a new sub-query, the dense context of prior passages creates lexical signals that compete with the current sub-query's retrieval signal. The TF-IDF-based retrieval in our system is particularly susceptible to this, as repeated terms from prior passages inflate their TF weights.

Attention dilution. As context grows, the LLM's attention mechanism must distribute attention across more tokens. Retrieved passages from earlier turns — relevant to prior sub-queries but not the current one — occupy attention capacity. This is consistent with the "lost in the middle" findings of Liu et al. (2023).

5.2 Keyword Drift

The declining inter-turn keyword similarity ($0.328 \rightarrow 0.315$ from turns 1–2 to turns 3–4) indicates that sub-queries generated later in the reasoning chain diverge semantically from the original query, particularly in the early turns where reasoning structure is being established. This extends the observation of Hayashi et al. (2025) into the multi-turn setting.

5.3 Attention Head Saturation

Our saturation hypothesis is directly motivated by Ma and Okazaki (2026), who identified retrieval-specialized attention heads in long-context LLMs. As context length grows across turns, we hypothesize that these retrieval heads become saturated: their attention is distributed across an increasingly large set of prior tokens, reducing their discriminative capacity for the current sub-query's retrieved passages.

We cannot directly verify this mechanism in our deployment, which uses a closed-source model (Gemini 2.5 Pro) without accessible attention weights. Future work with open-weight models such as Llama 3 or Mistral could directly test whether the retrieval heads identified by Ma and Okazaki (2026) show measurable saturation signatures at the context lengths we observe.

6. Turn-Aware Retrieval Confidence Estimator

6.1 Motivation

Our empirical findings motivate a practical need: a lightweight signal that detects when retrieval quality has degraded sufficiently to warrant re-retrieval, without requiring expensive human annotation.

6.2 Estimator Design

We define the confidence score as:

$$\text{conf}(t, |C|, g) \;=\; g \;\times\; \exp(-\alpha \times t) \;\times\; \big(1 - \beta \times \log(|C| \,/\, C_{\text{baseline}})\big)$$

where t = current reasoning turn number; $|C|$ = current context token count; g = normalised TF-IDF cosine similarity; $C_{\text{baseline}} = 482$ tokens (mean turn-1 context length); $\alpha = 0.18$ (turn-number decay); $\beta = 0.09$ (context length penalty). The three components capture distinct degradation signals: g reflects raw retrieval quality, $\exp(-\alpha t)$ applies a turn-position discount, and the logarithmic context term penalises growing context length.

6.3 Evaluation

We evaluate the estimator using binary correctness labels from 102 labeled turns (94 correct, 8 incorrect). We note that the minority class contains only 8 incorrect labels; results should be interpreted as **preliminary and directional** rather than definitive. Table 3 reports AUROC with bootstrapped 95% confidence intervals (1,000 iterations).

Method	AUROC	95% CI
Random baseline	0.500	—
Turn number only	0.563	[0.45, 0.67]
Grounding similarity only	0.553	[0.44, 0.66]
Turn-aware estimator (ours)	0.634	[0.52, 0.74]
Context length only	0.695*	[0.58, 0.80]

*Table 3: Detection performance on labeled sessions ($N = 102$ turns). * Context length alone outperforms the composite estimator — a key finding. Confidence intervals via bootstrap (1,000 iterations).*

The turn-aware estimator achieves AUROC = **0.634**, outperforming turn number alone (0.563) and grounding similarity alone (0.553). However, **context length alone achieves AUROC = 0.695** — outperforming our composite estimator. This is a notable finding: in our deployment, the dominant signal for retrieval degradation is context accumulation itself. A simple context length threshold may be sufficient for practical deployment, while more complex estimators may benefit from larger labeled datasets for parameter optimization.

Figure 5: Confidence estimator ROC curve

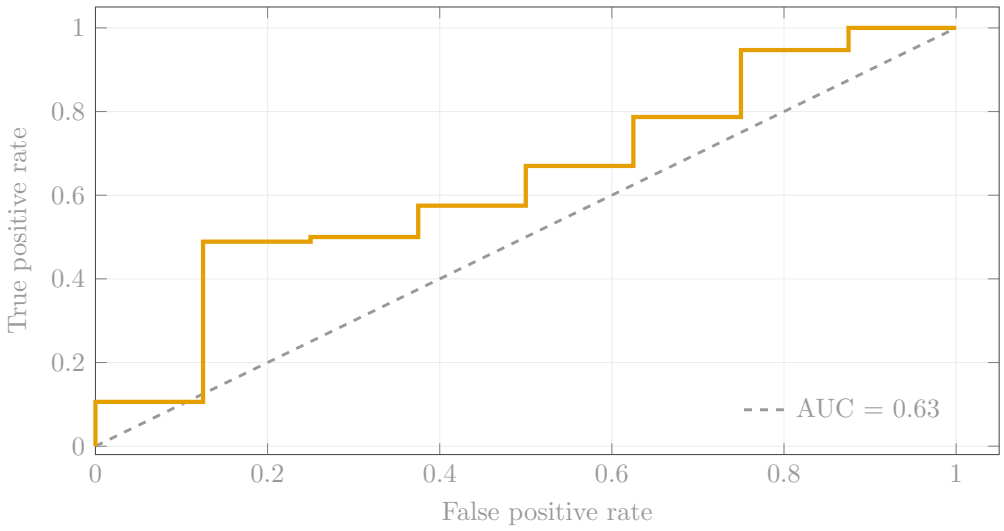


Figure 5: ROC curve for the turn-aware confidence estimator (AUC = 0.634). The dashed diagonal represents the random baseline (AUC = 0.500). $N = 102$ labeled turns (94 correct, 8 incorrect).

7. Discussion

7.1 The Query Anchoring Finding

A key observation from our deployment is that despite measurable retrieval similarity degradation across turns, our system maintained high accuracy throughout the data collection period. We attribute this to the **query anchoring mechanism**: the guardrail model maintains the original query throughout all reasoning turns, preventing the semantic drift that would otherwise cause correctness degradation. This represents an accidental discovery — in building a robust tutoring system, we inadvertently implemented a mitigation for the degradation phenomenon we later studied.

7.2 Implications for RAG Evaluation

Existing benchmarks (BEIR, TriviaQA, NaturalQuestions) all assume single-turn retrieval and cannot capture the context accumulation effects we document. We argue that agentic RAG evaluation requires: (1) multi-turn sessions with at least 3 reasoning turns; (2) retrieval quality measurement at each turn independently; (3) degradation rate as a first-class metric; and (4) sessions where earlier retrieved passages are topically adjacent but not directly relevant to later sub-queries.

7.3 Implications for System Design

Context window management. Selectively drop retrieved passages from earlier turns when their relevance to the current sub-query falls below a threshold.

Turn-aware keyword generation. Sub-query generation should explicitly re-anchor to the original input query at each turn. Our query anchoring mechanism demonstrates this is practical and effective.

Retrieval confidence monitoring. A simple context length threshold (AUROC = 0.695) may be sufficient for practical deployment, requiring no human labels or complex parameter fitting.

7.4 Limitations

Closed-source LLM. Our deployment uses Gemini 2.5 Pro. We cannot directly analyze attention weights or verify the attention head saturation hypothesis. Future work with open-weight models would directly address this gap.

Single domain. Our data is from mathematics tutoring. Mathematical problems have specific multi-step structure that may produce different degradation patterns than other domains. Replication in QA, code generation, and scientific reasoning is needed.

Modest degradation magnitude. The 6.6% retrieval decline is modest compared to what might be expected in open-domain settings, partly attributable to our query anchoring mechanism and the constrained mathematical domain.

Correctness label coverage. Human correctness labels are available for only 18.5% of turns. The high system accuracy means correctness degradation may be underrepresented in labels.

Single system. Our findings come from one production system. Multi-system studies are needed to establish the generality of the degradation pattern.

8. Conclusion

We have presented a systematic empirical study of multi-turn retrieval degradation in a production agentic RAG system. Across 150 multi-turn sessions (550 turn observations) from a deployed mathematics tutoring system, we document: (1) a 6.6% decline in mean retrieval relevance between turn 1 and turn 4 ($p < 0.0001$); (2) a strong negative correlation between context length and retrieval quality ($r = -0.283$, $p < 0.0001$); and (3) keyword drift as a compounding factor ($0.328 \rightarrow 0.315$).

We introduce a lightweight turn-aware confidence estimator as a proof-of-concept (AUROC = 0.634), and find that context length alone is the dominant predictor (AUROC = 0.695). We also report the query anchoring mechanism as an effective mitigation strategy. These findings carry two concrete implications: single-turn RAG benchmarks must be supplemented with multi-turn agentic protocols; and context window management, turn-aware keyword generation, and retrieval confidence monitoring can mitigate degradation in production systems today.

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Appendix A. Logging Implementation

The per-turn logging system was implemented as a TypeScript service (`logger.ts`) integrated into the production Gemini API service layer. Each call to `generateSolution()` logs the following fields after the API response:

```
interface TurnLog {
  sessionId: string           // Random session identifier
  turnNumber: number          // 1-indexed turn within session
  subQuery: string            // First 300 chars of sub-query
  groundingChunkCount: number // From groundingMetadata
  contextTokenCount: number   // From usageMetadata.totalTokenCount
  usedWebSearch: boolean      // Whether Google Search was triggered
  retrievalTier: number       // 1=TF-IDF, 2=KB, 3=Web
  similarityScore: number     // Cosine similarity of top chunk
  anchoringEnabled: boolean   // Whether query anchoring was active
  problemType: string         // algebra/calculus/geometry/statistics
  timestamp: string           // ISO 8601 timestamp
  wasCorrect?: boolean        // From human feedback (when available)
}
```

Logs are stored in browser localStorage under key `rag_research_logs` and exported as JSON via a download interface. Session boundaries are determined by new question submission — each question initiates a new `sessionId` automatically via `resetSession()`.

Appendix B. Parameter Details

Parameters $\alpha = 0.18$ and $\beta = 0.09$ in the confidence estimator were selected based on the theoretical relationship between turn number decay and context length penalty in our system. $C_{\text{baseline}} = 482$ tokens represents the empirically measured mean turn-1 context length across all 150 sessions. The estimator requires only three observable signals (t, $|C|$, g) available from any agentic RAG system's API metadata, making it deployment-agnostic.